

When Does a Classifier Help an LLM?

Classifier-Guided Prompting and Hybrid Classifier–LLM Models for Credit-Default Prediction

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Abstract—We ask a simple question. When does a trained classifier help a large language model (LLM) predict credit-card default? And when does the LLM help the classifier? We test this on the *Default of Credit Card Clients* data. We use a fixed set of 2,000 held-out clients. The data are imbalanced (22% default). We report recall, precision, F1, and AUC, all with bootstrap confidence intervals. We find three things. First, a few-shot LLM (claude-sonnet-5) catches more defaulters than the fitted models. It has the best recall (0.47) and F1 (0.50). But it ranks clients worse (AUC-ROC 0.72 vs. 0.79 for a Random Forest). Second, telling the LLM to imitate a classifier does not help. Saying “act as a Random Forest” or listing important features changes nothing. But *using* the classifier to prune the prompt to its top-8 features does help. Recall rises by 0.071 and F1 by 0.032, both significant. Curating the few-shot examples hurts. Third, the best method is a hybrid. We put the Random Forest’s probability into the prompt. This lifts the LLM’s AUC to 0.78, matching the Random Forest, while keeping higher recall (+0.118 over it). The reverse (LLM output into the classifier) does not help. Adding more classifiers does not help. The recipe is simple. Prune the prompt to a classifier’s important features. Feed one strong classifier’s probability into the LLM. Do not rely on imitation prompts, example curation, or the reverse combination.

I. INTRODUCTION

Credit-risk assessment is a canonical supervised classification task: given a client’s repayment history, credit limit, billing and payment amounts, and demographics, a model estimates the probability of default. Logistic regression and tree ensembles such as Random Forests are long-standing, strong baselines because they capture the nonlinear interactions typical of credit data [1]–[3].

Large Language Models (LLMs) have attracted interest for financial decision support because they follow structured instructions and produce natural-language explanations [4]. However, an LLM presented with tabular records differs fundamentally from a *fitted* classifier: a trained classifier learns parameters from labeled data, whereas an LLM can only be *instructed* at inference time. Prior work has noted the difficulty of applying LLMs directly to numerical/tabular prediction and has generally kept prediction grounded in a conventional model while using the LLM for explanation [5]–[7].

This raises a practical question that, to our knowledge, has not been carefully isolated for credit risk: *when, and how, does a fitted classifier actually help an LLM (and vice versa)?* We distinguish three ways a classifier’s knowledge can be transferred to an LLM prompt and test each in a controlled manner:

- 1) **Imitation instructions:** tell the LLM to reason like the classifier (role and/or feature-importance hints).
- 2) **Classifier-guided prompt construction:** use the classifier to decide *what to put in the prompt*—which features to show, and which few-shot examples to use.
- 3) **Hybrid composition:** feed the classifier’s output (predicted probability and decision) into the prompt as input, or conversely feed the LLM’s output into the classifier.

A. Related Work

LLMs for credit and finance. A growing body of work applies LLMs to credit and financial tasks. Feng et al. [8] instruction-tune a credit LLM (CALM) and benchmark it against GPT-4; Sanz-Guerrero and Arroyo [9] build a default-risk indicator from peer-to-peer loan descriptions. Domain models and surveys map the broader landscape [10], [11], and a systematic review of interpretable credit LLMs catalogs architectures and explanation mechanisms [12]. Most of this work targets *textual* inputs (loan descriptions, disclosures) or fine-tuned domain models. We instead study *prompting over structured tabular records* with a general model, and how a fitted classifier can assist it.

LLMs for tabular prediction. TabLLM [13] shows that few-shot LLM classification of serialized tables is competitive with trees at very low shot counts; a survey [14] and newer methods extend this through fine-tuning [15] and retrieval-augmented in-context learning [16]. These works establish that LLMs *can* predict on tabular data but generally find tree ensembles stronger as data grow—consistent with our ranking (AUC) results.

Combining classifiers and LLMs. Closest to us, Huetas [17] compares gradient-boosted trees and LLMs for few-

shot tabular learning. A parallel line uses LLMs to *select* or *engineer* features: LLM-Select [18] and Li et al. [19] perform prompt-based feature selection, and FeatLLM [20] engineers features for a downstream model. Our feature-pruning condition inverts this idea: we use a *fitted* classifier’s importances to prune the prompt, and we further test injecting the classifier’s *output* into the prompt (a hybrid) in both directions.

Class imbalance and synthetic data. SMOTE [21] remains the standard minority oversampler; diffusion models such as TabDDPM [22] and LLM-based synthesizers (EPIC [23]) generate tabular data, including for imbalance. Our augmentation study compares a diffusion generator against these standard remedies for credit default.

B. Contributions and Differences from Prior Work

Prior work shows LLMs can predict on tables and that classifiers can guide LLMs via feature selection; it does not settle *how* to combine a fitted classifier with an LLM for imbalanced credit risk, nor which transfer mechanism actually works. We make this precise:

- **Controlled, imbalance-aware comparison.** Unlike accuracy-focused or text-based credit-LLM studies [8], [9], [12], we compare a few-shot LLM against fitted logistic regression, Random Forest, and AdaBoost on a fixed imbalanced set using per-class recall/F1 and AUC-ROC/PR with paired bootstrap confidence intervals, reporting null and negative results explicitly.
- **Imitation vs. construction.** We separate *telling* the LLM to imitate a classifier from *using* the classifier to build the prompt. Imitation (role and/or feature hints) gives no significant gain; classifier-guided *feature pruning* does. This contrasts with LLM-driven feature selection [18]–[20], which selects features *with* an LLM rather than pruning the prompt *with* a fitted model.
- **Example curation is harmful.** Selecting few-shot examples by decision-boundary proximity or prototypicality significantly *hurts*; random balanced examples are better—a cautionary result for exemplar selection.
- **Bidirectional hybrids.** Beyond tree-vs-LLM comparison [17], we test both composition directions. Feeding the classifier’s probability into the prompt gives Random-Forest-level ranking with significantly higher recall; the reverse (LLM output into the classifier) and richer ensembles do not help.
- **Augmentation baseline.** We compare diffusion-based minority augmentation [22] against SMOTE and class-weighting with multi-seed confidence intervals for credit default.

II. DATASET

We use the *Default of Credit Card Clients* dataset [1]: 30,000 clients, 23 features (credit limit, sex, education, marital status, age; repayment-status codes PAY_0–PAY_6; bill amounts BILL_AMT1–6; payment amounts PAY_AMT1–6), and a binary default label. The data are imbalanced: 22.1% default.

We split 80/20 with a fixed seed, *stratified* by label (24,000 train / 6,000 test). Fitted models train on the full 24,000. Because LLM inference is comparatively expensive, all LLM conditions are evaluated on a fixed, stratified subset of 2,000 held-out test clients (default rate 22.1%); every LLM condition sees the same 2,000 clients. The diffusion-augmentation study (Section IV-D), which involves only fitted models, is evaluated on the full 6,000-client test set.

III. METHODOLOGY

A. Fitted baselines

Majority predicts the majority (no-default) class. **Logistic Regression** is trained on standardized features. **Random Forest** (RF) uses 100 trees, maximum depth 12. **AdaBoost** uses 200 estimators. All use a fixed seed.

B. LLM prediction

The LLM is `claude-sonnet-5` with extended thinking disabled for near-determinism. Each client’s features are serialized as text; the model is given $k=12$ balanced labeled few-shot examples (6 default, 6 non-default) drawn from the training set and asked to return, for each client, a JSON object with a binary `label` and a `confidence` $\in [0, 1]$ interpreted as $P(\text{default})$ (used for AUC). Requests are batched (20 clients per call) and cached. We evaluate the following LLM conditions.

Imitation prompting.

- **L1 (direct):** predict from features and the few-shot examples, no classifier guidance.
- **L2 (full):** “act as a trained Random Forest” *and* a list of the RF’s top features to weigh.
- **L2a (role only):** “act as a trained Random Forest,” no feature list.
- **L2b (features only):** weigh the RF’s important features, no role.

Classifier-guided prompt construction (instruction text held constant at the plain L1 wording; only prompt *content* changes).

- **E1 (top- K features):** include only the RF’s top-8 features; drop the other 15.
- **E2 (boundary shots):** choose the 12 few-shot examples nearest the RF decision boundary (out-of-bag probability closest to 0.5), balanced by class.
- **E3:** E1 + boundary shots.
- **E4 (representative shots):** choose the most confidently-correct examples (out-of-bag probability near 1 for defaulters, near 0 for non-defaulters), balanced.
- **E5:** E1 + representative shots.

Few-shot verdicts used for selection come from out-of-bag/cross-validated probabilities so they are not overfit.

Hybrid composition.

- **E6 (RF $\rightarrow$ LLM):** E1 + the RF’s predicted probability and 0/1 decision as extra inputs for each client (and each few-shot example, using out-of-bag verdicts).

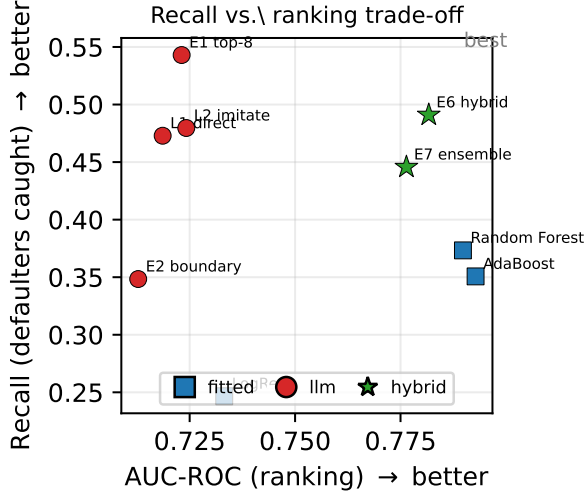


Fig. 1. Recall vs. AUC-ROC across models (N=2000). Fitted models (squares) rank well but catch few defaulters; plain/prompt-engineered LLM conditions (circles) catch more defaulters but rank worse; the hybrid E6 (star) attains Random-Forest-level ranking and high recall.

- **E7 (ensemble → LLM):** E1 + the probabilities and decisions of *three* classifiers (RF, logistic regression, AdaBoost).
- **Inverse (LLM → RF):** add the LLM’s E1 label and confidence as features to the RF; evaluated by 5-fold cross-validation on the 2,000 (comparing RF with vs. without the LLM feature on identical folds).

C. Metrics and uncertainty

We report accuracy, and—because the data are imbalanced—precision, recall, and F1 on the default class, plus AUC-ROC and AUC-PR. Overall accuracy is reported alongside the majority-baseline accuracy (0.779) to contextualize it. Uncertainty on the 2,000-client set uses 1,000 bootstrap resamples; comparisons between conditions use *paired* bootstrap (same resampled indices), and we call a difference significant when its 95% CI excludes zero. The diffusion study reports mean $\pm$ 95% CI across 5 seeds.

IV. RESULTS

Figure 1 previews the central trade-off: fitted models occupy the high-AUC/low-recall region, LLM conditions the low-AUC/high-recall region, and the hybrid E6 reaches the desirable high-AUC/high-recall corner.

A. LLM vs. fitted models, and imitation prompting

Table I reports all fitted baselines and the imitation-prompting LLM conditions on the 2,000-client set.

Two observations. (i) The LLM *trades precision for recall*: the direct LLM (L1) flags $\sim 20\%$ of clients as defaulters vs. the RF’s 12%, giving the highest recall (0.47) and, with E1, the highest F1 (0.53) of any model—above the fitted RF (0.48)—while the fitted models retain higher precision, accuracy, and

TABLE I
FITTED BASELINES AND IMITATION-PROMPTING LLM CONDITIONS (N=2000, DEFAULT RATE 22.1%). BEST VALUE PER COLUMN IN BOLD.

Model	Acc	Prec	Rec	F1	AUROC	AUPRC
Majority	0.779	—	0.000	0.000	—	—
Logistic Reg.	0.811	0.708	0.247	0.366	0.733	0.512
Random Forest	0.821	0.668	0.373	0.479	0.790	0.577
AdaBoost	0.823	0.698	0.351	0.467	0.793	0.571
L1: direct LLM	0.789	0.525	0.473	0.498	0.719	0.481
L2: role+features	0.790	0.526	0.480	0.502	0.724	0.478
L2a: role only	0.795	0.540	0.477	0.507	0.722	0.487
L2b: features only	0.790	0.526	0.489	0.506	0.728	0.486
E1: top-8 feats	0.786	0.515	0.543	0.529	0.723	0.470

TABLE II
PAIRED DIFFERENCES VS. L1 (MEAN [95% CI], 1000 BOOTSTRAP RESAMPLES). BOLD = SIGNIFICANT (CI EXCLUDES 0).

Condition	$\Delta F1$	$\Delta AUROC$	$\Delta Recall$
E1 top-8 feats	+0.032 [0.006, 0.056]	+0.005 [−.013, 0.022]	+0.071 [0.040, 0.102]
E2 boundary shots	−0.057 [−.088, −.025]	−0.006 [−.018, 0.006]	−0.125 [−.158, −.092]
E3 top-8+boundary	−0.011 [−.039, 0.015]	+0.006 [−.013, 0.023]	−0.045 [−.075, −.015]
E4 repr. shots	−0.029 [−.054, −.006]	+0.001 [−.009, 0.013]	−0.070 [−.097, −.044]
E5 top-8+repr.	+0.020 [−.005, 0.046]	+0.011 [−.004, 0.027]	+0.016 [−.012, 0.047]
E6 (E1+RF)	+0.026 [−.000, 0.054]	+0.063 [0.039, 0.085]	+0.019 [−.010, 0.048]
E7 (E1+RF+LR+ADA)	+0.026 [†]	+0.058 [†]	−0.010 [†]

[†]E7 differences vs. L1 shown for reference; E7 is compared against E6 in the text.

AUC. The LLM is thus recall-competitive but ranking-inferior. (ii) **Imitation prompting does not help**: paired bootstrap differences of L2, L2a, and L2b versus L1 are non-significant on F1, AUC-ROC, and recall (all 95% CIs include zero; e.g., L2–L1 F1 = +0.004, CI [−0.013, +0.020]). Neither the “act as a Random Forest” role nor an explicit feature-importance list changes performance.

B. Classifier-guided prompt construction

Table II reports paired differences versus L1 for the prompt-construction and hybrid conditions.

Feature pruning helps; example curation hurts. Restricting the prompt to the RF’s top-8 features (E1) significantly improves recall (+0.071) and F1 (+0.032)—and E1 attains the best F1 of any model in the study (0.529), above the fitted RF. In contrast, both example-curation strategies significantly *reduce* recall and F1: boundary (hardest-case) selection (E2) is worst ($\Delta F1$ −0.057, $\Delta recall$ −0.125), and representative (prototypical) selection (E4) also significantly hurts ($\Delta F1$ −0.029). Combining feature pruning with curated examples (E3, E5) dilutes E1’s gain to non-significance. The takeaway is that a classifier is useful for deciding *which features* the LLM should see, but random balanced few-shot examples are better than RF-curated ones.

C. Hybrid classifier–LLM composition

RF → LLM (E6) is the best operating point. Adding the RF’s probability and decision to the prompt raises AUC-ROC from 0.719 (L1) to 0.782—a +0.063 significant gain that essentially matches the fitted RF (0.790)—and AUC-PR from

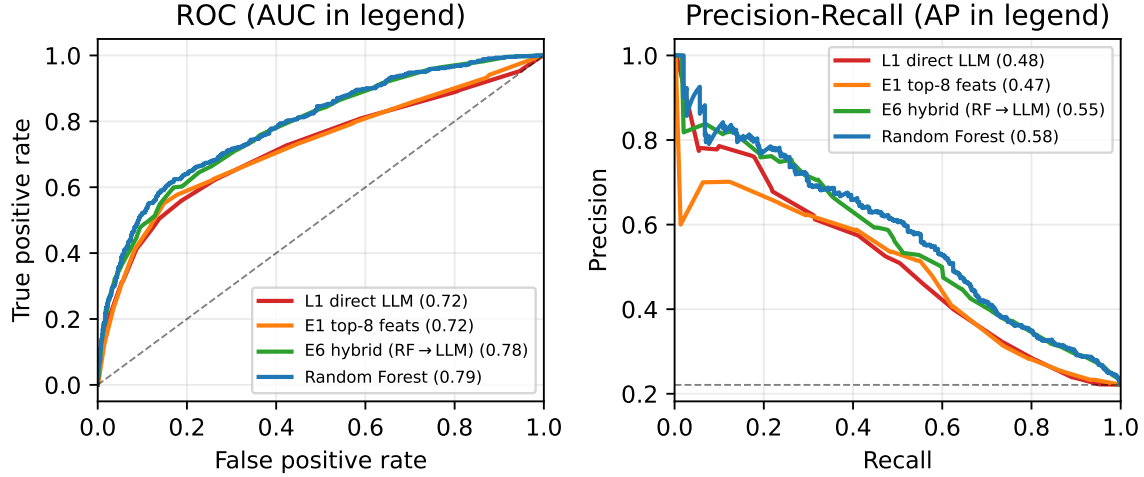


Fig. 2. ROC (left) and precision-recall (right) curves on the 2000-client test set for the direct LLM (L1), feature-pruned LLM (E1), the hybrid (E6), and the fitted Random Forest. Feeding the RF’s probability into the prompt (E6) lifts the LLM’s ROC/PR curves to essentially the Random Forest’s level.

TABLE III

HYBRID AND REFERENCE MODELS (N=2000). E6/E7 vs. RF PAIRED: E6 RECALL +0.118 (SIG), F1 +0.045 (SIG), AUROC −0.008 (SIG). E7 vs. E6: RECALL −0.045 (SIG), F1/AUROC N.S.

Model	Acc	Prec	Rec	F1	AUROC	AUPRC
E1 (feats only)	0.786	0.515	0.543	0.529	0.723	0.470
E6 (E1+RF)	0.803	0.561	0.491	0.524	0.782	0.549
E7 (E1+RF+LR+ADA)	0.811	0.595	0.446	0.510	0.776	0.536
Random Forest	0.821	0.668	0.373	0.479	0.790	0.577

0.481 to 0.549. Crucially, E6 retains high recall: versus the fitted RF, E6 improves recall by +0.118 and F1 by +0.045 (both significant), at a negligible AUC cost (−0.008). Thus E6 attains Random-Forest-level ranking with substantially higher recall (Table III).

More classifiers do not help. Feeding three classifier opinions (RF, logistic regression, AdaBoost; E7) does not improve ranking over the RF-only hybrid (E7 vs. E6: Δ AUC-ROC −0.005, n.s.; Δ F1 −0.014, n.s.) and significantly *reduces* recall (−0.045). The three classifiers are highly correlated (all AUC-ROC $\approx$ 0.79, all high-precision/low-recall), so their combined signal is redundant and pushes the LLM toward a more conservative consensus.

The reverse composition does not help. Adding the LLM’s E1 output (label and confidence) as features to the RF, evaluated by 5-fold cross-validation on the 2,000 clients, yields no significant change: RF-with-LLM versus RF-without on identical folds gives Δ recall +0.002, Δ F1 −0.004, Δ AUC-ROC −0.002 (all n.s.). Notably the RF ranks the LLM’s confidence as its most important input, yet metrics do not move—the LLM’s signal, being derived from the same features, is redundant with what the RF already extracts. Information useful for the hybrid flows one way: the classifier supplies ranking ability the LLM lacks, but the LLM supplies nothing

TABLE IV

RANDOM FOREST UNDER IMBALANCE REMEDIES (FULL TEST, N=6000; MEAN $\pm$ 95% CI OVER 5 SEEDS).

Strategy	Recall	F1	AUPRC
original	0.351 $\pm$.004	0.458 $\pm$.003	0.556 $\pm$.003
class_weight	0.575 $\pm$.004	0.538 $\pm$.003	0.550 $\pm$.003
oversample	0.561 $\pm$.005	0.537 $\pm$.003	0.547 $\pm$.001
SMOTE	0.541 $\pm$.006	0.511 $\pm$.005	0.518 $\pm$.004
diffusion	0.541 $\pm$.045	0.526 $\pm$.017	0.540 $\pm$.015

the classifier lacks.

D. Diffusion-based minority augmentation

Because recall is limited by class imbalance, we test whether augmenting the training data with *synthetic defaulters* from a diffusion model helps a fitted RF, compared with standard remedies. We train a Gaussian denoising diffusion model (with a quantile-normal transform of continuous features) on the defaulter rows and generate synthetic defaulters to rebalance the training set; a detection classifier distinguishes real from synthetic defaulters with AUC 0.80 (imperfect but usable). Table IV reports RF performance on the full 6,000-client test set, averaged over 5 seeds.

Paired across seeds, diffusion augmentation *significantly* beats SMOTE—the standard synthetic-minority baseline—on AUC-PR (+0.022), precision (+0.028), and accuracy (+0.013), while tying on recall and F1. However, it does *not* beat simple class-weighting, which is free and gives the highest recall/F1, and diffusion shows higher run-to-run variance. Diffusion augmentation is therefore a better synthetic generator than SMOTE for this task but is not worth its complexity relative to class-weighting.

V. DISCUSSION

Our results give a consistent picture of how a fitted classifier and an LLM should—and should not—be combined for imbalanced tabular credit prediction.

Telling an LLM to imitate a classifier is ineffective, but using the classifier to shape the prompt is effective in one specific way: selecting which features to show. This is intuitive—dropping 15 low-importance columns reduces distraction—and it is the only prompt-only manipulation that significantly helped. Curating few-shot examples, by either difficulty or prototypicality, hurt; random balanced examples were consistently better, suggesting that example *diversity* matters more than example “informativeness” for this task.

The strongest practical result is the hybrid: feeding a single strong classifier’s probability into the prompt gives the LLM the ranking ability it otherwise lacks (AUC-ROC 0.72 $\rightarrow$ 0.78) while preserving its higher recall—yielding a model that ranks like the Random Forest but catches substantially more defaulters. Because Random Forest inference is essentially free, this hybrid is cheap to deploy. The asymmetry is informative: the reverse composition (LLM $\rightarrow$ classifier) and richer ensembles (three classifiers $\rightarrow$ LLM) added nothing, indicating that a single strong classifier already supplies the complementary signal the LLM needs.

VI. LIMITATIONS

Results use a single LLM (claude-sonnet-5), a single dataset, one 2,000-client evaluation draw, and one few-shot draw per condition. Bootstrap CIs capture test-set sampling uncertainty but not LLM-sampling or few-shot-draw variance; the ablation and hybrid conclusions would be strengthened by multiple prompt draws and additional models. The inverse-composition test trains the cross-validated RF on $\sim 1,600$ rows per fold, below the 24,000-row RF, though the RF-with vs. RF-without comparison is controlled. The LLM used only the same features available to the classifiers; an LLM with access to information the classifier lacks (e.g., unstructured text) might contribute complementary signal, which we did not test. The diffusion generator treats ordinal repayment codes as continuous; a fully mixed-type generator might improve synthetic quality.

VII. CONCLUSION

We studied how to combine a fitted classifier and an LLM for credit-default prediction. A few-shot LLM is good at recall but weak at ranking. Telling the LLM to imitate a classifier does not help. Using the classifier to choose which features go in the prompt does help. Curating the few-shot examples does not help.

The best model is a hybrid. We give the LLM the Random Forest’s probability. It then ranks like the Random Forest and still catches more defaulters. Random Forest inference is almost free, so this hybrid is cheap. The reverse direction does not help. Adding more classifiers does not help. Separately, diffusion-based augmentation beats SMOTE but not simple class-weighting.

Our advice is simple. Prune the prompt to a classifier’s important features. Add one strong classifier’s probability to the prompt. Do not rely on imitation prompts, example curation, or the reverse combination.

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APPENDIX A

PROMPT AND REPRODUCIBILITY DETAILS

All clients were serialized as comma-separated name=value fields. The base prompt states the task, briefly defines the feature groups (PAY_* repayment codes, BILL_AMT, PAY_AMT, LIMIT_BAL), lists $k=12$ balanced labeled examples, gives the condition-specific instruction, lists the clients, and requires a JSON array of `{id, label, confidence}` with no prose. In E1/E3/E5 the serialization is restricted to the RF’s top-8 features (PAY_0, PAY_2, BILL_AMT1, PAY_4, PAY_3, LIMIT_BAL, PAY_AMT1, BILL_AMT2); in E6/E7 each row additionally carries classifier probabilities and 0/1 decisions (out-of-bag/cross-validated for few-shot examples, fitted-model outputs for test clients). Data split: stratified 80/20, seed 42; LLM evaluation on a stratified 2,000-client subset of the test set. Fitted models: RF (100 trees, depth 12), logistic regression (standardized), AdaBoost (200 estimators). LLM: `claude-sonnet-5`, extended thinking disabled, batched at 20 clients per request with response caching. Uncertainty: 1,000 bootstrap resamples (paired for between-condition comparisons); diffusion study over 5 seeds. Diffusion generator: Gaussian DDPM (cosine schedule) on quantile-normal-transformed defaulter rows.

Reproducibility. All code, data-preparation scripts, per-condition prompts, cached LLM outputs, result tables, and the scripts that produce every figure are available at https://github.com/researchsparkhub/credit_risk_llm (made public upon publication). Because the LLM responses are cached in the repository, every reported number can be regenerated without new API calls; the dataset is rebuilt from the public UCI source via the included conversion script. Random seeds are fixed (42; diffusion confidence intervals use seeds 0–4).